

Binary and bases

How can computers solve complex calculations when all they understand is electrical signals? Binary numbers are used to translate these signals into numbers.

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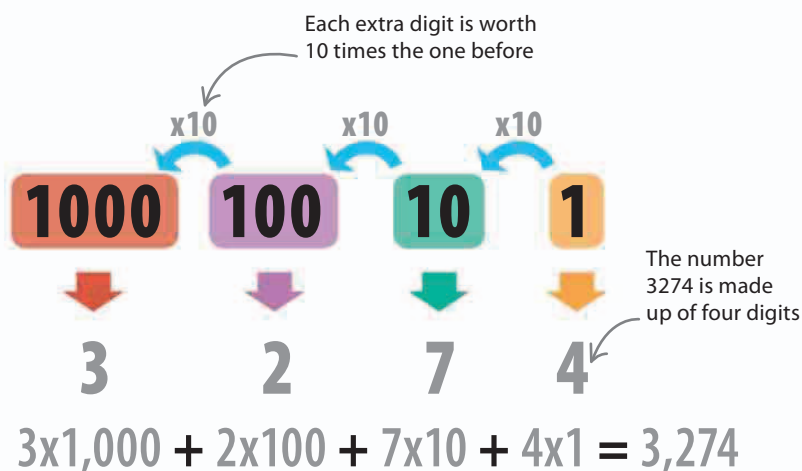
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What is a base number?

A “base” is the number of values that can be shown using only one digit. Each extra digit increases the number of values that can be shown by a multiple of the base.

▷ Decimal system

The decimal system is the most familiar counting system, and has a base of 10. It can show 10 values with one digit, 100 values with two digits, and 1,000 with three digits.



Binary code

At the most basic level, computers only understand two values: electrical signals that are “on” and “off”. Because there are only two values, computers deal with numbers using a base of two, or “binary”. Each digit is either a 1 or a 0, and each extra digit in the number is worth two times the previous digit.

